

SEVEN ROW-COLUMN FACTORIAL DESIGNS OF STRENGTH TWO AT $m = 6$ LISTED AS UNKNOWN BY RAHIM AND CAVENAGH

ABSTRACT. Rahim and Cavenagh conjectured that a Hadamard matrix of order $4m$ forces the existence of an $I_k(4m, 4n, 2, 2)$ whenever $n \geq m$, $2^k \mid 16mn$ and $k \leq 4m - 1$, with the single exception $m = 1$ and n odd. They verified the conjecture for $m \leq 5$ and recorded that at $m = 6$ the unknown cases with minimal parameters are exactly $I_{k+3}(24, 2^k, 2, 2)$ for $5 \leq k \leq 11$. We exhibit one witness that settles all seven: a 24×23 orthogonal array M of strength 2 from the Paley Hadamard matrix of order 24, and a fixed 3-dimensional binary code D of minimum weight 3, such that $A[i][j] = g_i + v_j$ is an $I_{k+3}(24, 2^k, 2, 2)$, where the g_i are the rows of the first $k+3$ columns of M and the v_j run over $D^\perp$. Both objects are printed in full. The reading of the source's definition under which these are the objects listed as unknown is the one fixed by the source's own necessary conditions and by the examples it prints. The same witness reproduces the authors' already published range $12 \leq k \leq 20$; nothing here settles the case $m = 6$ of the conjecture in full.

1. THE CONJECTURE

Fix integers $q, t \geq 2$ and $k \geq t$. An $\text{OA}(N, k, q, t)$ is an $N \times k$ array over $[q]$ in which every $N \times t$ subarray contains each of the q^t words of $[q]^t$ exactly N/q^t times. Following Rahim and Cavenagh [4, Section 1], an $I_k(m, n, q, t)$ is an $m \times n$ array whose mn cells are filled with the λ -fold repetition of every word of $[q]^k$, where $\lambda = mn/q^k$, in such a way that, on fixing any t of the k coordinates, each of the q^t patterns occurs exactly n/q^t times in every row and exactly m/q^t times in every column. Equivalently, every row is an $\text{OA}(n, k, q, t)$ and every column is an $\text{OA}(m, k, q, t)$. We only need $q = t = 2$, where the condition on a pair of coordinates $\{a, b\}$ is that each of 00, 01, 10, 11 occurs $n/4$ times in every row and $m/4$ times in every column.

The sentence that introduces this definition, in the version of [4] we read and identically in [3, p. 79], calls the array formed by a row (column) an orthogonal array “of size k , degree n (respectively, m)”, which interchanges the two slots of the source's own $\text{OA}(N, k, q, t)$ notation: a row of an $m \times n$ array of words of length k holds n such words, so as an array it is $n \times k$. Three statements of the source fix the reading, and all three agree with the definition just given. First, the sentence immediately following it says that fixing t coordinates inside one row (column) leaves a full factorial design with

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replication n/q^t (respectively m/q^t). Second, the source's own necessary conditions read: if an array of type $I_k(m, n, q, t)$ exists then $q^k \mid mn$ and there exist an $OA(m, k, q, t)$ and an $OA(n, k, q, t)$ [3, p. 81], with k in the degree slot of both. Third, the one printed example whose shape can settle the orientation is printed with m rows: its $I_5(12, 8, 2, 2)$ appears as 12 rows of 8 words, the twelve rows marked by the twelve superscripts $A, \dots, L$ [3, Table 5.3]. The dimensions of the printed $I_4(9, 9, 3, 2)$, $I_4(12, 12, 2, 2)$ and $I_6(12, 16, 2, 2)$ agree with their labels as well, though the first two are square and the third is set sideways, so none of them distinguishes the orientation [3, Tables 5.1, 5.5, 5.6]. Section 3 says which of this is machine checked.

The statement we settle cases of is Conjecture 5.48 of the first author's thesis [3, p. 114], which appears also as a displayed conjecture in [4]. We quote it from the thesis together with the paragraph that follows it there, which is what makes the present note finite.

Conjecture 5.48. If there exists a Hadamard matrix of order $4m$, then there exists an $I_k(4m, 4n, 2, 2)$ for any $n \geq m$, $2^k \mid 16mn$ and $k \leq 4m - 1$, with the exception $m = 1$ and n is odd.

From Theorem 5.39, the above conjecture is true for $m \leq 5$. For $m = 6$, the unknown cases with minimal parameters are: $I_{k+3}(24, 2^k, 2, 2)$; $5 \leq k \leq 11$. From Theorem 5.36 and the existence of a Hadamard matrix of order 12, $I_{k+3}(24, 2^k, 2, 2)$ exists for $12 \leq k \leq 20$.

Theorems 5.39 and 5.36 are earlier results of the same chapter; the numbering in the published version of [4] differs.

So the source's list of what is unknown at $m = 6$ with minimal parameters has exactly seven members, and Theorem 1 below produces all seven at once. Each is an instance of the conjecture: with $4m = 24$ and $4n = 2^k$ one has $m = 6$, $n = 2^{k-2} \geq 8 > 6$ for $k \geq 5$, $2^{k+3} \mid 16mn = 3 \cdot 2^{k+3}$, and $k + 3 \leq 14 \leq 4m - 1 = 23$, while $m = 6 \neq 1$; and a Hadamard matrix of order 24 exists, one being written down in Section 2.

2. THE WITNESS

Let $q = 23$ and let χ be the quadratic-residue character modulo 23, so $\chi(0) = 0$ and $\chi(a) = \pm 1$ according as $a \neq 0$ is or is not a square. Index rows and columns of a 24×24 matrix by $0, 1, \dots, 23$ and set $C_{0,0} = 0$, $C_{0,j} = 1$ and $C_{i,0} = -1$ for $i, j \geq 1$, and $C_{i,j} = \chi(j - i)$ for $i, j \geq 1$. Then $H = C + I$ is a Hadamard matrix of order 24 [2]. Negate those rows of H whose entry in column 0 is -1 , delete column 0, and replace -1 by 0. The result is the 24×23 binary array

g_0	11111111111111111111111111111111	g_{12}	001101011111000001010011
g_1	000001010011001101011111	g_{13}	10011010111100000101001
g_2	100000101001100110101111	g_{14}	11001101011110000010100
g_3	110000010100110011010111	g_{15}	01100110101111000001010
g_4	111000001010011001101011	g_{16}	00110011010111100000101
g_5	111100000101001100110101	g_{17}	10011001101011110000010
g_6	011110000010100110011011	g_{18}	01001100110101111000001
g_7	101111000001010011001101	g_{19}	10100110011010111100000
g_8	010111100000101001100111	g_{20}	01010011001101011110000
g_9	101011110000010100110011	g_{21}	00101001100110101111000
g_{10}	110101111000001010011001	g_{22}	00010100110011010111100
g_{11}	011010111100000101001101	g_{23}	00001010011001101011110

which we call M . Throughout, a word of $\mathbb{F}_2^K$ is a string of K binary digits whose position t , counted from the left starting at 0, is coordinate t ; addition is coordinatewise modulo 2; and $\langle x, y \rangle$ is the modulo-2 inner product. It is a standard consequence of $HH^T = 24I$ that M is an $\text{OA}(24, 23, 2, 2)$ [1, p. 148]: every one of the $\binom{23}{2} = 253$ coordinate pairs sees each of 00, 01, 10, 11 in exactly 6 of the 24 rows. For $8 \leq K \leq 23$ write $G^{(K)}$ for the $24 \times K$ array of the first K columns of M , whose rows we call $g_0^{(K)}, \dots, g_{23}^{(K)}$; deleting coordinates preserves strength 2, so each $G^{(K)}$ is an $\text{OA}(24, K, 2, 2)$.

The second ingredient is fixed once and for all. Let

$$h_1 = 11100000, \quad h_2 = 01011000, \quad h_3 = 10010100,$$

regarded inside $\mathbb{F}_2^K$ by padding with zeros, and let $D = \langle h_1, h_2, h_3 \rangle$. The three are independent, so $|D| = 8$, and the seven nonzero words of D have weights 3, 3, 3, 3, 4, 4, 4; in particular

- (1) every nonzero word of D has weight at least 3.

Let $V^{(K)} = D^\perp \leq \mathbb{F}_2^K$, a subspace of dimension $K - 3$ and hence of size 2^{K-3} . For $K = 8$ its 32 words are

00000000	11010000	01101000	10111000
10100100	01110100	11001100	00011100
00000010	11010010	01101010	10111010
10100110	01110110	11001110	00011110
00000001	11010001	01101001	10111001
10100101	01110101	11001101	00011101
00000011	11010011	01101011	10111011
10100111	01110111	11001111	00011111

(read left to right, top to bottom; the order is immaterial in Theorem 1). Finally, let $\sigma: \mathbb{F}_2^K \rightarrow \mathbb{F}_2^3$ be the syndrome map $\sigma(x) = (\langle h_1, x \rangle, \langle h_2, x \rangle, \langle h_3, x \rangle)$. Its fibres are the eight cosets of $V^{(K)}$, and since h_1, h_2, h_3 are supported on the first eight coordinates, σ does not depend on K . On the 24 rows of M it takes each of the eight values of $\mathbb{F}_2^3$ exactly three times:

- (2) $|\{i : \sigma(g_i) = s\}| = 3$ for every $s \in \mathbb{F}_2^3$,

equivalently, each of the seven nonzero $h \in D$ satisfies $\langle h, g_i \rangle = 1$ for exactly 12 of the 24 rows. Both (1) and (2) are finite statements about the digits printed above.

Theorem 1. *Let $8 \leq K \leq 23$, let $g_0^{(K)}, \dots, g_{23}^{(K)}$ be the rows of $G^{(K)}$, let $v_0, \dots, v_{2^{K-3}-1}$ be the words of $V^{(K)} = D^\perp$ in any order, and put*

$$A[i][j] = g_i^{(K)} + v_j \quad (0 \leq i \leq 23, 0 \leq j \leq 2^{K-3} - 1).$$

Then A is an $I_K(24, 2^{K-3}, 2, 2)$ of index 3. Consequently $I_{k+3}(24, 2^k, 2, 2)$ exists for every k with $5 \leq k \leq 20$; in particular for the seven values $5 \leq k \leq 11$ listed as unknown in the conjecture quoted above.

Proof. Write $n = 2^{K-3}$ and $V = V^{(K)}$, $g_i = g_i^{(K)}$, so that A is $24 \times n$ and $24n = 3 \cdot 2^K$.

Multiplicity. Row i of A lists the coset $g_i + V$, each of its n elements once. For $x \in \mathbb{F}_2^K$ we have $x \in g_i + V$ if and only if $\sigma(x) = \sigma(g_i)$, because the fibres of σ are the cosets of $V = \ker \sigma$. Hence x occurs in exactly as many rows as there are i with $\sigma(g_i) = \sigma(x)$, which is 3 by (2). So every word of $\mathbb{F}_2^K$ occurs exactly $\lambda = 3 = 24n/2^K$ times in A .

Rows. Fix i and a pair of coordinates $a \neq b$. As v runs over V , the entry $g_i + v$ has (a, b) -pattern $((g_i)_a, (g_i)_b) + (v_a, v_b)$, so the four pattern counts of row i are those of the linear map $\pi: V \rightarrow \mathbb{F}_2^2$, $\pi(v) = (v_a, v_b)$, permuted by a translation of $\mathbb{F}_2^2$. Now π is surjective unless some nonzero $\alpha e_a + \beta e_b$ ($\alpha, \beta \in \mathbb{F}_2$) annihilates V , that is, lies in $V^\perp = D$; and every such word has weight 1 or 2, which (1) forbids. So π is onto, its fibres are cosets of $\ker \pi$ and each has $|V|/4 = n/4$ elements. Every pattern therefore occurs $n/4$ times in row i .

Columns. Fix j . Column j of A is $\{g_i + v_j : 0 \leq i \leq 23\}$, that is $G^{(K)}$ with the single constant vector v_j added to every row. Adding a constant permutes the four patterns of any coordinate pair among themselves, so the pattern counts of column j equal those of $G^{(K)}$, namely $6 = 24/4$ each, because $G^{(K)}$ is an $OA(24, K, 2, 2)$.

Thus A is an $I_K(24, n, 2, 2)$ with $n = 2^{K-3}$ and $\lambda = 3$. Writing $K = k + 3$ turns this into $I_{k+3}(24, 2^k, 2, 2)$, and $8 \leq K \leq 23$ is $5 \leq k \leq 20$. $\square$

The proof is the specialisation to $\dim D = 3$ of a mechanism of [4], reproduced here so that the note is self-contained; no originality is claimed for it, nor for the range $12 \leq k \leq 20$, which the quoted passage itself records as following from a published theorem of [4] together with a Hadamard matrix of order 12. What was missing was a pair (M, D) satisfying (1) and (2) at $4m = 24$. The seven cases $5 \leq k \leq 11$ are those the quoted passage lists as unknown, and we claim nothing beyond that listing. Nothing above proves the case $m = 6$ of the conjecture, which asserts an $I_k(24, 4n, 2, 2)$ for every admissible pair (k, n) with $n \geq 6$; the remaining parameters, in particular $4n \in \{4b, 8b, 16b\}$ with b odd, are untouched here, no case $m \geq 7$ is settled,

and no nonexistence is proved. We make no claim that the witness above is unique, canonical, or least in any sense.

On the literature: a search of the arXiv listings for row-column factorial designs and of the authors' own arXiv postings, made in August 2026, found nothing later than [4] bearing on these seven cases, and the thesis passage quoted above still states them as unknown. We make no stronger claim about the literature than that.

3. WHAT WAS CHECKED

Every input of Theorem 1 is printed in Section 2, so a reader can redo the argument without a computer. The accompanying program `verify.py` uses the Python standard library only and works in exact integer arithmetic. It rebuilds H from $q = 23$, checks all 576 inner products of its rows and confirms that normalising and reducing H gives the 24×23 array printed above digit for digit; checks that M is an $\text{OA}(24, 23, 2, 2)$; checks (1) and (2), and that the 32 printed words of $V^{(8)}$ are exactly $D^\perp$; checks the hypotheses of Theorem 1 for all sixteen column counts $8 \leq K \leq 23$; and verifies each of the seven designs $5 \leq k \leq 11$ cell by cell against the definition of Section 1, 1 363 104 multiplicity and pattern counts in all. The recorded run `verify.output.txt` reports **VERDICT: ALL 32 CHECKS PASS** and exit status 0; the checks it reports beyond those just listed concern statements this note does not make.

Three of the checks concern the reading of the definition. The label and the printed dimensions of each of the four designs the source exhibits are transcribed into the program, and it checks that the dimensions match the label, with the rows of the one non-square example the source marks as rows being the m of that label. It then checks that the reading of Section 1 makes the replication numbers n/q^t , m/q^t and the index mn/q^k positive integers for all four of those labels and for the seven labels at issue here, while the reading that takes the interchanged slots literally — so that the array formed by a row would have k rows and n columns rather than n rows and k columns — would require $n = k$, which fails for all eleven. Finally it exhibits, for each of the seven, the two orthogonal arrays that the source's own necessary conditions demand: the first K columns of M as an $\text{OA}(24, K, 2, 2)$ and $V^{(K)}$ as an $\text{OA}(2^{K-3}, K, 2, 2)$.

Three limits are worth stating. The cells $12 \leq k \leq 20$ are not checked against the definition, since their arrays have up to 2^{20} columns; for those only the hypotheses of Theorem 1 are checked, which is what the theorem consumes. The digits of the source's own printed examples are not transcribed and no run of them is replayed here; only their dimensions and their parameter arithmetic enter the checks described above. And the passages of the source quoted or paraphrased in Section 1 were read by us and are not machine checkable at all.

REFERENCES

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- [3] F. Rahim, *Row-column factorial designs and mutually orthogonal frequency rectangles*, Ph.D. thesis, University of Waikato, 2023. hdl:10289/15893. The conjecture quoted in Section 1 is Conjecture 5.48, on p. 114. The definition is on p. 79, the necessary conditions on p. 81, and the four printed examples in Tables 5.1, 5.3, 5.5 and 5.6, on pp. 80, 104, 106 and 115.
- [4] F. Rahim and N. J. Cavenagh, *Row-column factorial designs with strength at least 2*, Linear Algebra Appl. (2023), doi:10.1016/j.laa.2023.02.018. A version of this paper is Chapter 5 of the thesis above. The version of it we read is arXiv:2207.02397v3; we quote from the thesis, and the printed journal wording and numbering may differ from both.